

Quantitative Portfolio Optimization Report

Construction and evaluation of a maximum Sharpe ratio equity portfolio using Modern Portfolio Theory.

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1. Methodology

The portfolio was constructed using Modern Portfolio Theory. Historical stock returns were used to estimate expected returns and covariance. Portfolio weights were optimized by maximizing the Sharpe ratio. The resulting portfolio was evaluated through return, volatility, risk-adjusted performance measures and drawdown analysis. A benchmark portfolio was created using an equal-weight allocation across the same 17 stocks to determine whether optimization added value.

2. Portfolio Performance

Metric	Value
Annual Return	15.93%
Annual Volatility	21.10%
Sharpe Ratio	0.755
Sortino Ratio	1.045
VaR 95%	-1.92%
CVaR 95%	-2.93%
Maximum Drawdown	-34.56%
Calmar Ratio	0.461

3. Benchmark Comparison

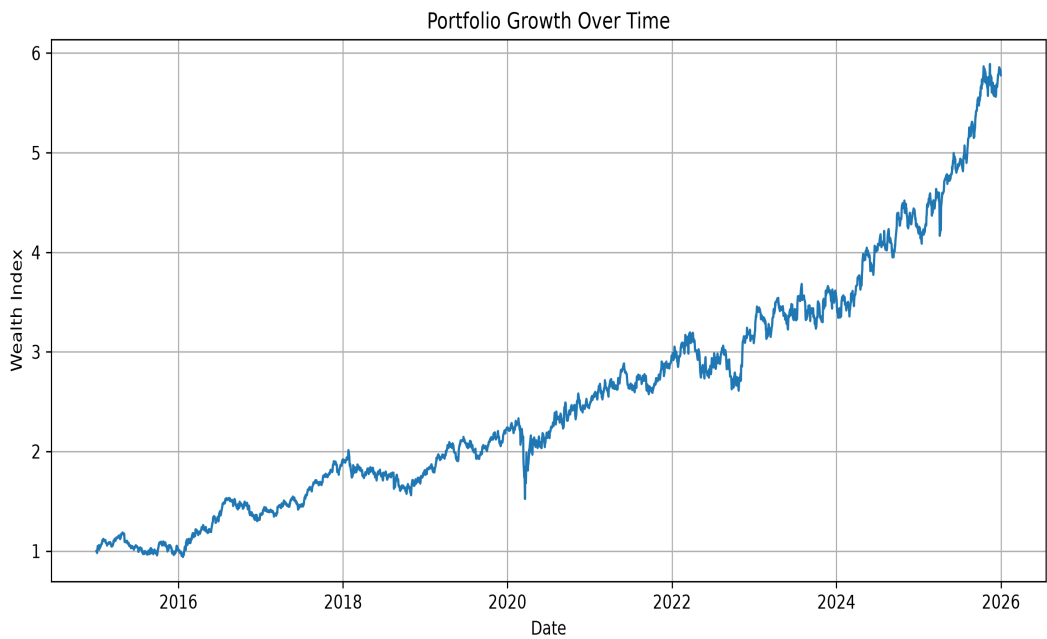
The optimized portfolio is compared against an equal-weight benchmark consisting of the same securities. This comparison evaluates whether portfolio optimization improved risk-adjusted returns.

Metric	Optimized Portfolio	Equal Weight Benchmark
Annual Return	0.1592662738233252	1.6961822661001376
Annual Volatility	0.2110333717753933	3.923266109906249
Sharpe Ratio	0.754697100669155	0.4323393363038201
Sortino Ratio	1.0449700392404544	10.397490211442047
Maximum Drawdown	-0.3618590561287556	-0.5079203354807652
Calmar Ratio	0.4401334473349633	3.339465163359996

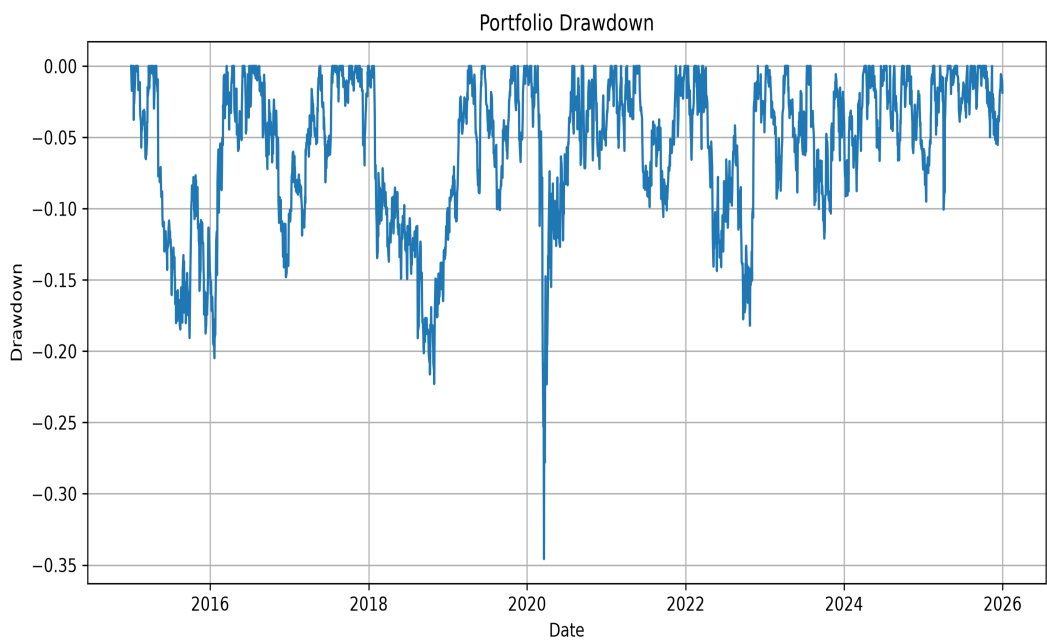
The results reveal a clear trade-off between stability and raw performance. The benchmark delivers a much higher annual return, yet this comes at the cost of extreme volatility. Its annual volatility is nearly twenty times greater than that of the optimized portfolio, making its returns highly unstable and unpredictable. By contrast, the optimized portfolio achieves far lower volatility and smaller drawdowns, offering a more consistent and resilient performance profile. Risk-adjusted measures further highlight this distinction. The optimized portfolio records a stronger Sharpe ratio, indicating that it generates more return per unit of risk than the benchmark. This suggests that, despite lower absolute returns, the optimized portfolio is more efficient in balancing reward against risk. Interestingly, the benchmark shows unusually high Sortino and Calmar ratios, which may reflect calculation quirks or the disproportionate influence of its outsized returns relative to downside risk. Nevertheless, these figures do not negate the fact that the benchmark's volatility and drawdowns are significantly larger. Taken together, the comparison underscores the essence of portfolio

optimization. It is not about chasing the highest possible return, but about achieving a sustainable balance between growth and risk. The optimized portfolio demonstrates that stability, resilience, and efficiency can outweigh raw performance, especially for investors who prioritize long-term risk management over short-term gains.

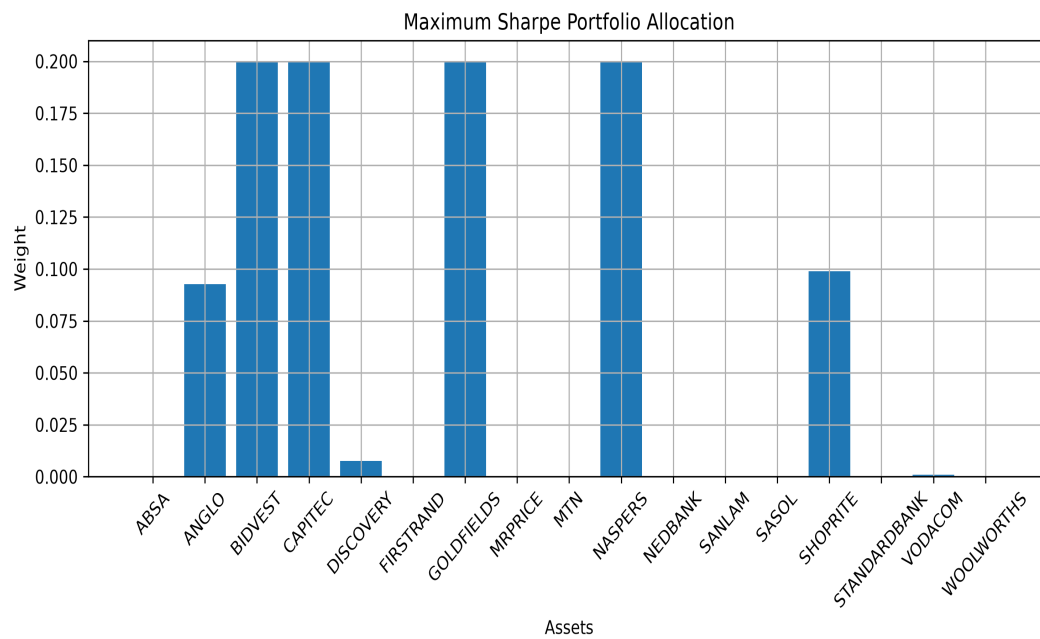
4. Portfolio Growth



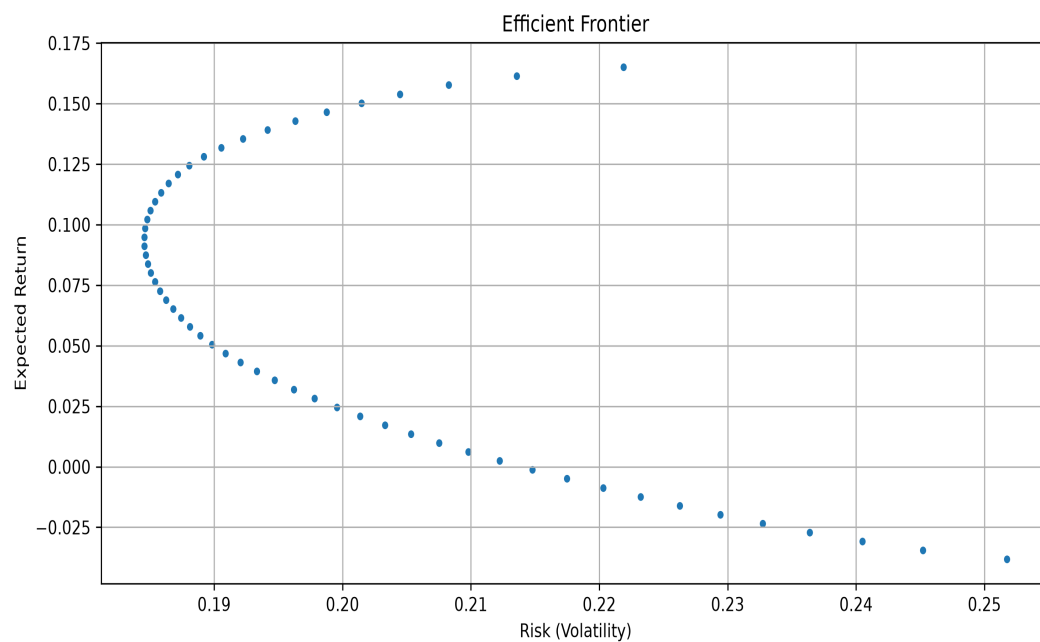
5. Drawdown Analysis



6. Portfolio Allocation



7. Efficient Frontier



8. Conclusion

This study demonstrated the application of quantitative portfolio optimization techniques in constructing and evaluating an equity investment portfolio. Using historical return data from seventeen South African listed companies, a maximum Sharpe ratio portfolio was developed based on Modern Portfolio Theory principles, where asset weights were optimized to achieve the most efficient risk-adjusted return profile. The empirical results indicate that the optimized portfolio provided an improved balance between return and risk compared with an equally weighted benchmark portfolio. By incorporating the historical relationships between asset returns through covariance estimation, the optimized portfolio was able to achieve a more efficient allocation of capital across the selected securities. The comparison with the equal-weight benchmark suggests that systematic portfolio construction and diversification techniques can enhance investment outcomes by reducing unnecessary exposure to individual securities while maintaining attractive return potential. From a wealth management perspective, the findings highlight the importance of disciplined asset allocation rather than relying solely on individual security selection. The optimized portfolio demonstrates how quantitative methods can assist investment managers in making objective allocation decisions, managing portfolio risk and improving risk-adjusted performance. Measures such as the Sharpe ratio, Sortino ratio and maximum drawdown provide evidence that portfolio evaluation should consider both return generation and downside risk management. However, the results should be interpreted within the limitations of the methodology. Portfolio optimization relies heavily on historical estimates of expected returns and correlations, which may not remain stable in changing market environments. Market conditions, economic cycles, liquidity constraints and investor objectives can significantly influence future portfolio performance. Therefore, the optimized portfolio should not be viewed as a guaranteed superior investment strategy, but rather as a quantitative framework that supports informed portfolio decision-making. Overall, the analysis confirms that quantitative optimization techniques can provide valuable tools for modern wealth management by enabling investors to construct portfolios that are aligned with their desired risk-return objectives. Combining mathematical optimization with professional judgement and continuous portfolio monitoring remains essential for achieving sustainable long-term investment outcomes.